

Reverse Osmosis

When two solutions of unequal concentration are separated by a semi-permeable membrane, which does not permit the passage of dissolved solid particles (molecules and ions), the flow of solvent takes place from dilute solution to concentrated solution. This process is called osmosis.

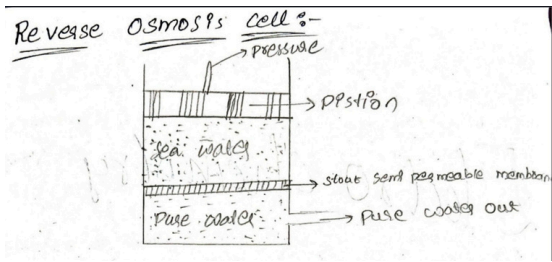
If a hydrostatic pressure greater than osmotic pressure is applied on the concentrated side, the solvent is forced to move from higher concentration to lower concentration side across the membrane.

Thus, the solvent flow is reversed. Hence, this method is called reverse osmosis.

Thus, in reverse osmosis, pure water is separated from contaminated water.

This membrane filtration is called super filtration (or) hyper filtration.

Reverse Osmosis Cell



(Pressure is applied on salt water side → semi-permeable membrane → pure water comes out)

Advantages

- Cost of purification of water is less and maintenance cost is also less
- This water can be used for high pressure boilers
- Both ionic and non-ionic, colloidal, high molecular weight organic matter is removed from the water sample

Calomel Electrode

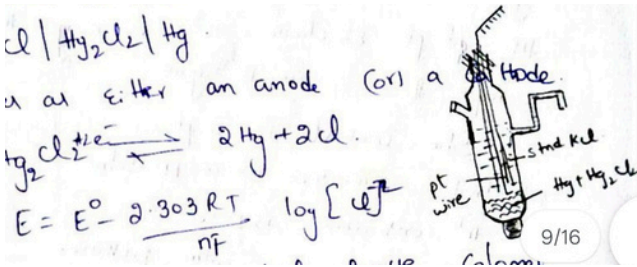
It consists of a glass tube at the bottom of which there is a layer of mercury (Hg). Over this is placed a paste of mercury and mercurous chloride (Hg₂Cl₂). The remaining portion of the tube is filled with a solution of potassium chloride (KCl).

A platinum (Pt) wire dipping into the mercury layer is used for making electrical contact. The calomel electrode is represented as:



It can act either as an anode or cathode.

The electrode reaction is: Hg₂Cl₂ ⇌ 2Hg + Cl₂



The electrode potential of the calomel electrode depends on the concentration of KCl.

Determination of pH of Unknown Solution using SHE and Calomel Electrode

To determine the pH of an unknown solution, a Standard Hydrogen Electrode (SHE) is coupled with a calomel electrode to form an electrochemical cell. The potential difference between these electrodes is measured using a potentiometer, which is then used to calculate the pH.

Procedure / Setup

1. The SHE consists of a platinum (Pt) electrode immersed in a solution of H⁺ ions with hydrogen gas (H₂) bubbling at 1 atm around it.
2. The calomel electrode is a reference electrode containing Hg, Hg₂Cl₂ (calomel), and saturated KCl solution.
3. These two electrodes are connected through a salt bridge.
4. The unknown solution is placed in contact with the SHE.

5. The calomel electrode serves as the reference electrode, providing a stable and known potential.

6. The saturated calomel electrode is connected to the SHE via a potentiometer, and the unknown solution is tested.

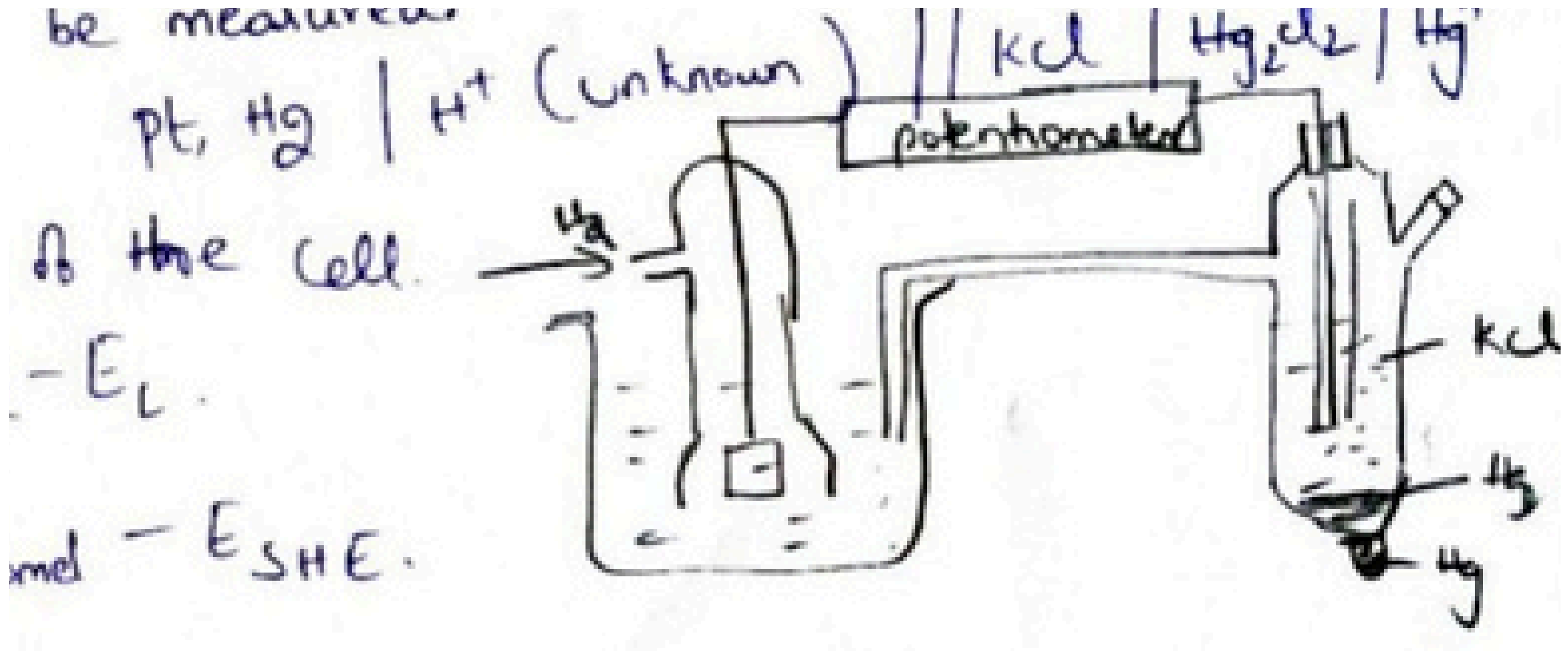
Cell Representation

$\text{Pt, H}_2 \mid \text{H}^+ (\text{unknown}) \parallel \text{KCl} \mid \text{Hg}_2\text{Cl}_2 \mid \text{Hg}$

EMF of the Cell

$E_{\text{cell}} = E_R - E_L$

$E_{\text{cell}} = E_{\text{calomel}} - E_{\text{SHE}}$



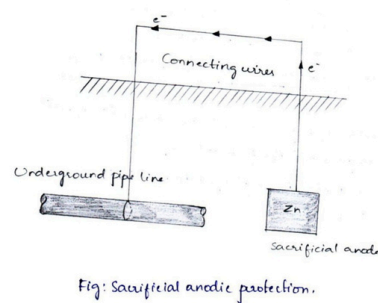
Cathodic Protection

The principle involved in cathodic protection is to force the metal to behave like a cathode.

The important cathodic protection methods are:

1. Sacrificial anode protection
 2. Impressed current cathodic protection
- 1) Sacrificial Anode Protection Method

In this method, the metallic structure to be protected is made cathode by connecting it with a more active metal.



So that all the corrosion will concentrate only on the active metal. The original metal (structure) gets protected, while the active metal gradually gets corroded. Hence, this process is otherwise known as sacrificial anodic protection.

Metals like aluminium, zinc, magnesium are used as sacrificial anodes.

Applications of Sacrificial Anodic Protection

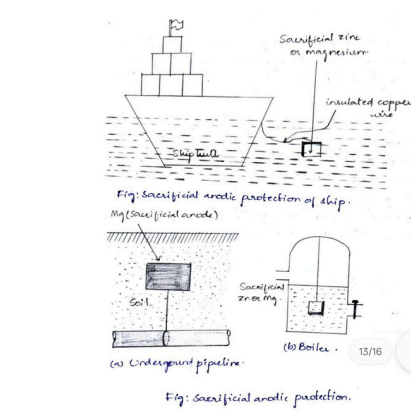
a) Protection of ships and boats

Sheets of Mg and Zn are hung around the hull of the ship. Zn/Mg act as anodes compared to iron, so corrosion concentrates on Zn/Mg.

b) Protection of underground pipelines and cables from corrosion

c) Insertion of Mg sheet into domestic water heaters to prevent rust formation

d) Calcium metal is used to minimize engine corrosion



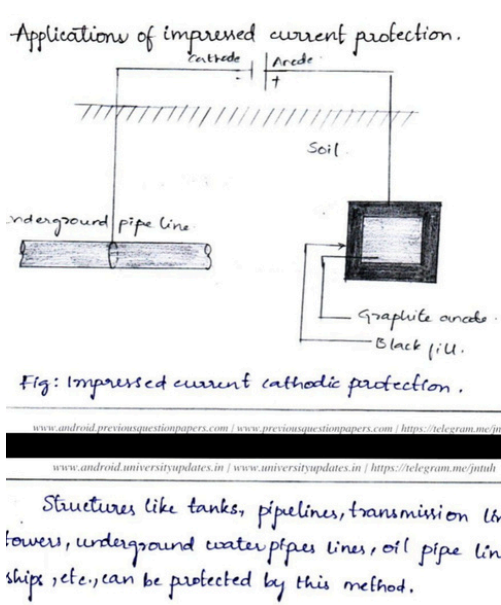
2) Impressed Current Cathodic Protection Method

In this method, an impressed current is applied in opposite direction of corrosion current to nullify it, converting the corroding metal into cathode.

This is done by connecting the negative terminal of a battery to the metal structure to be protected and the positive terminal to an inert anode (graphite/platinized titanium).

The anode is buried in backfill (like coke breeze) to provide good electrical contact.

Applications of Impressed Current Protection



- Storage tanks
- Pipelines
- Transmission lines
- Underground structures like gas and pipe lines

Construction, Working and Applications of Zn–Air Battery

Construction

A Zn–air battery (ZAFB) consists of two electrodes: a zinc (Zn) anode and an air cathode.

The anode and cathode are separated by a separator which allows ions to transfer across the cell.

Potassium hydroxide (KOH) aqueous solution is used as an electrolyte.

Working

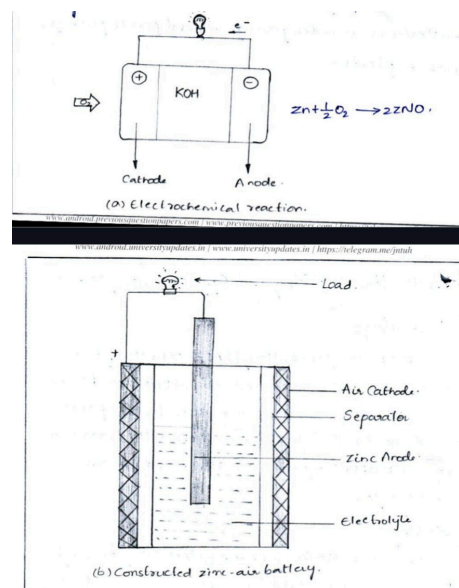
Zn–air batteries generate electricity through the electrochemical reaction of zinc and oxygen.

During discharge:

- Zinc anode is oxidized and produces zincate, which later changes to zinc oxide
- At the cathode, oxygen from the atmosphere undergoes reduction

Overall reaction:





Applications of Zn–Air Battery

The important applications of zinc–air batteries are:

- Used as power sources for hearing aids
- Used in medical devices
- Used in pagers
- Used as temporary/disposable power packs for mobile phones

Chlorination

The process of adding chlorine to the water to kill the bacteria is called chlorination.

There are 3 types of chlorinating agents are used: Adding chloramines

Adding bleaching powder

Adding chlorine (Cl₂)

Adding chloramines

When chlorine is mixed in ammonia solution, it reacts to form chloramines. It further reacts with water to form hypochlorous acid which acts as a powerful disinfectant.



HOCl acts as disinfectant.

Bleaching powder

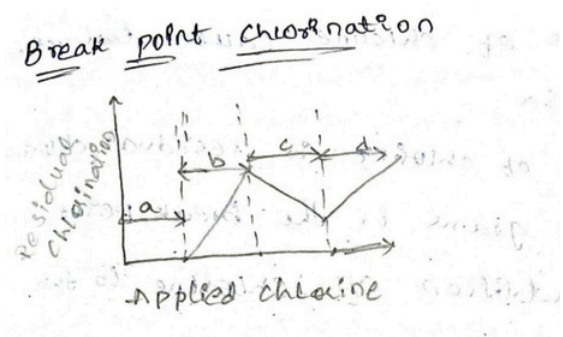
Adding bleaching powder to the water releases hypochlorous acid which acts as a powerful disinfectant.



Chlorination

The process of chlorine to kill the bacteria present in water is called chlorination.

Break point chlorination



a → oxidizes reducing agents

b → forms chloramines

c → destroys bacteria

d → forms residual chlorine

The water sample treated with chlorine and estimated for residual chlorine in water and plotted a graph as shown above which gives break point chlorination.

From the graph it is clear that:

1. a grams of chlorine added oxidizes reducing agents
2. b grams of chlorine forms chloro compounds (amines)

Desalination of water

Water containing high concentration of dissolved salts and having a brackish taste is called brackish water.

Sea water is an example of brackish water containing about 3.5% dissolved salts.

This water cannot be used for domestic & industrial purposes. The dissolved salts are removed by desalination.

Ion-exchange Methods

De-ionisation:

It involves exchange of cation of hardness causing salts with H⁺ ions or ions of hardness causing salts with OH⁻ ions. For this purpose, two types of ion exchange resins are used:

1. Cation exchange resin
2. Anion exchange resin

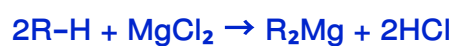
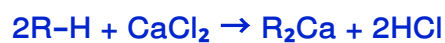
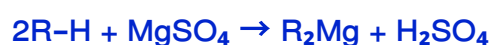
Principle:

Ion-exchange resins are insoluble long chain organic polymers and the functional groups attached to the chain can exchange hardness producing cations and anions present in water.

Process:

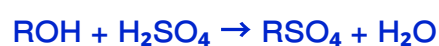
Step 1:

The hard water is passed through cation exchange resin where calcium & magnesium ions are exchanged with H⁺ ions of the resin.



Step 2:

The water is then passed through anion exchange resin where chloride, sulphate & nitrate ions are exchanged with OH⁻ ions present in the resin.



Thus, this process removes all types of hardness producing cations & anions present in water. The resulting water is known as demineralised (or) deionised water.

Regeneration of Resins:

With constant use, the resins get exhausted and can be regenerated as follows:

- Exhausted cation resin can be regenerated by passing dilute HCl



Calcium chloride & magnesium chloride are removed as wash.

- Exhausted anion resin can be regenerated by passing sodium hydroxide



NaCl & sodium bicarbonate are removed as wash.

